

Harish Venkatesan M R

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PERSONAL PROFILE

Robotics engineer with an MSc in Intelligent Robotics from the University of York and 18 months of commercial software engineering experience. Hands-on background in autonomous navigation, multi-sensor integration, embedded control and real-time computer vision, built through robotic systems delivered from concept to working demonstration. Seeking a robotics, automation or controls engineering role in the UK.

TECHNICAL SKILLS

- Robotics and Autonomy - autonomous navigation, path planning, obstacle avoidance, sensor fusion, real-time control loops, robot kinematics and dynamics, Mecanum wheel drive systems, ROS 2 (Nav2, SLAM Toolbox, Gazebo)
- Control and Instrumentation - feedback control, closed-loop tuning, sensor-actuator synchronisation, instrumentation for robotic systems
- Sensors and Embedded - IR, ultrasonic and LiDAR integration, camera systems, Arduino microcontroller programming, serial communication protocols, embedded systems design
- Computer Vision and Machine Learning - YOLOv5 object detection, OpenCV image processing, real-time video pipelines, applied machine learning
- CAD and Simulation - Autodesk Inventor, Fusion 360, MATLAB, Simulink
- Programming and Tools - Python, C/C++ (Arduino), JavaScript, React.js, Git version control; basic familiarity with Linux systems and conventions
- Engineering Practice - system integration, hardware-software debugging, test and validation, technical documentation, agile delivery

ROBOTICS PROJECTS

Autonomous Search and Rescue System

University of York, 2024

Team Lead, team of 4 - integrated aerial and ground robotic system for disaster response

- Led a team of 4 engineers, coordinating subsystem ownership across the ground and aerial platforms and integrating workstreams into a single working system.
- Delivered autonomous navigation for a ground robot, integrating IR and ultrasonic sensors for obstacle detection, with movement control programmed on Arduino, control logic in Python, and Mecanum wheels for omnidirectional movement in confined spaces.
- Built a real-time object detection pipeline using YOLOv5 to identify victims and obstacles from live camera input, feeding detections directly into navigation decisions.
- Built the serial communication layer linking the sensor array to a custom-built two-jaw gripper, enabling the robot to autonomously grasp a detected rescue object and return it to the start position without operator input.
- Resolved sensor-motor synchronisation faults across the communication chain that were causing unreliable motion and inconsistent gripper actuation.

Autonomous Waypoint Navigation Robot

- Built a wheeled mobile robot that navigates autonomously between two defined waypoints and returns to its origin without operator input.
- Implemented sensor fusion combining IR and ultrasonic range data to improve detection reliability, compensating for IR sensitivity to surface reflectivity and ultrasonic errors on angled or soft surfaces.
- Developed reactive collision avoidance enabling the robot to detect obstacles, manoeuvre around them and resume its route to the target waypoint.

EXPERIENCE

Associate Engineer - Technology, Virtusa

May 2022 - September 2023

Chennai, India

- Developed and maintained production application features using JavaScript and React.js in a client-facing consultancy environment, working to defined delivery standards and release deadlines.
- Participated in code reviews, refactoring activities and continuous improvement initiatives across the engineering team.
- Developed and executed unit and integration testing activities to ensure software reliability and maintainability.
- Supported production deployments, issue investigation, troubleshooting and release management, building disciplined fault-isolation habits directly applicable to hardware-software integration work.

Software Engineering Intern, Virtusa

January 2022 - March 2022

- Completed a structured engineering internship and progressed to a full-time Associate Engineer role.

EDUCATION

MSc Intelligent Robotics, University of York

September 2023 - January 2025

Modules - Robot Kinematics and Dynamics; Control and Instrumentation for Robotics; Practical Robotics; Embedded Systems Design and Programming; Machine Learning

BE Computer Science and Engineering, KPR Institute of Engineering and Technology

August 2018 - May 2022

First Class. Modules - Data Structures; Object Oriented Programming; Computer Networks; Human-Computer Interaction; NLP

RIGHT TO WORK

Graduate Route (Post-Study Work) visa - can work full-time in the UK without sponsorship until January 2027; seeking a Skilled Worker sponsor thereafter. Open to relocate.